Meta-Kaizen Series · Paper 1 of 5

Meta-Kaizen: A General Theory and Algorithmic Framework for the Mathematical Formalization of Self-Evolving Continuous Improvement Across Arbitrary Governance Substrates

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Series Orientation

This is Paper 1 of five in the Meta-Kaizen series. Paper 1 derives the general theory, axiomatic foundation, and scoring function. Paper 2 proves the networked implementation and governance closure theorems. Paper 3 extends the framework to reasoning propagation and institutional memory. Paper 4 integrates the Fractal Rhythm Model for regime-aware adaptive governance. Paper 5 provides the decision-theoretic capstone: the formal framework for when a rational agent should act on approaching system tipping points, completing the series. Each paper is designed to be self-contained for the reader with background in operations research, measurement theory, or organizational governance; the series introduction provides full context for readers entering at other points.

Abstract

The philosophy of Kaizen—continuous, incremental improvement—has been practiced across manufacturing, healthcare, software engineering, and public governance for nearly four decades since its systematic Western exposition (Imai, 1986). Yet despite this ubiquity, Kaizen has never been formally mathematized. No substrate-agnostic, algorithmically reproducible, self-referential framework exists for scoring, ranking, and governing improvement candidates across arbitrary structured processes or documents. The result is a persistent gap between the richness of Kaizen practice and the poverty of its theoretical foundations.

This paper addresses that gap. We introduce Meta-Kaizen: a formally specified, substrate-independent, algorithmically implementable framework for continuous self-improvement. The core contribution is the Kaizen Variation Score (KVS), a multiplicative priority function defined over four normalized, dimensionless components—Novelty (N), Impact (I′), inverse Complexity (C′), and Timeliness (T)—each rigorously bounded in [0,1] so that KVS ∈ [0,1]. The

multiplicative structure enforces non-substitutability: a zero in any dimension collapses the score, preventing high impact from compensating for zero novelty or zero timeliness.

We derive the multiplicative functional form axiomatically from six measurement-theoretic axioms (Luce & Tukey, 1964; Krantz et al., 1971), including a novel Essentialness with Veto Power axiom that formally justifies non-substitutability. The equal weighting of components is derived from a Marginal Symmetry axiom combined with a scale normalization condition, not stipulated. The common exponent $\alpha = 1$ is selected as the canonical choice by the parsimony principle that the scoring function should be linear in each normalized component at face value—a modeling principle, not a forced mathematical result. The adoption threshold $\kappa = 0.50$ is derived from Bayesian decision theory under symmetric losses, with the general asymmetric-case formula presented and its distributional assumptions made explicit.

Section 5 presents an honest structural simulation analysis: KVS achieves 73.8% recall [90% CI: 71.2, 76.2] against a noisy multiplicative ground truth—an earned result, not an analytic tautology. Sensitivity analyses across six distributional parameter variants and five top-k thresholds confirm structural robustness. The Timeliness dominance artifact ($\rho = 0.740$ with KVS) is explained and operationally remedied. Section 6 demonstrates the framework across four substrate domains, including a worked example that clears the adoption gate after refinement, illustrating KVS used both diagnostically and prescriptively.

A planned open-source Python library (metakaizen) is under development as a reference implementation with substrate-agnostic adapters, a CLI interface, a domain rotation scheduler, and a structured audit log. When released, the paper and library will together constitute a reproducible, extensible foundation for a formally verifiable implementation of Kaizen as a mathematical system. The library specification is described in Section 8; the repository will be published at GitHub under an open license prior to or concurrent with peer review.

Keywords: Kaizen formalization · continuous improvement theory · substrate-agnostic governance · multiplicative priority scoring · orthogonal inspiration · bisociation · novelty scoring · self-evolving systems · structural validation · antifragility · organizational learning

AI contributions: Claude (Anthropic) provided mathematical formalization, literature synthesis, axiomatic derivation review, and manuscript writing. Grok (xAI) provided empirical data processing and quantitative validation support. All general theory and mathematical formulations are contributed to the public domain.

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1. Introduction

Kaizen—the compound of the Japanese kai (change) and zen (good)—entered the Western management vocabulary through Imai's (1986) landmark systematization of Toyota's production philosophy. In the nearly four decades since, it has proliferated across domains that would have seemed alien to the Nagoya assembly lines where Taiichi Ohno first operationalized just-in-time production in 1948. Hospitals deploy it to reduce medication errors (Berwick, 1989; Langley et al., 2009). Software teams use it to compress deployment cycles (Forsgren, Humble, & Kim, 2018).

Public agencies apply it to administrative reform (Suárez-Barraza & Ramis-Pujol, 2010). By any measure, Kaizen is among the most successful management frameworks of the twentieth century.

Yet this success is built on a foundation that has never been formalized. The operations research literature treats Kaizen as a philosophy, a culture, or at most a set of heuristics—never as a mathematical object with defined properties, provable theorems, or algorithmic implementations. The practical consequence is that no two Kaizen implementations are comparable. A hospital's PDSA cycle and a semiconductor fab's kaizen event share a name and a spirit but no common measurement language. Whether an improvement was novel, well-timed, low-complexity, or high-impact is judged entirely by local intuition.

A second, deeper limitation follows: classical Kaizen cannot improve itself. The improvement engine has no self-referential structure. Meta-improvement—the application of the improvement method to the improvement method—requires a formal specification of what 'improvement' means in the abstract, independently of the substrate being improved. This is precisely what has been absent from the literature.

Referee note: We anticipate the immediate objection that mathematical formalization is unnecessary given Kaizen's demonstrated empirical success. The answer is yes, for three reasons. First, without formalization, implementations cannot be compared, replicated, or audited—Kaizen remains a craft rather than a science. Second, without a formal scoring function, improvement candidates are selected by authority or intuition, introducing systematic biases. Third, self-evolution is impossible without a mathematical object that can be turned back on itself. A philosophy cannot be an input to itself; an algorithm can.

This paper introduces Meta-Kaizen: a general mathematical theory and algorithmic framework that formalizes Kaizen as a substrate-agnostic, self-evolving process. The core contributions are:

(1) A formal definition of KVS with all components rigorously normalized to [0,1] and three formal properties proved: boundedness, monotonicity, and self-referential applicability.

(2) A complete axiomatic derivation of the multiplicative functional form from six axioms, including the derivation of equal weights and the pinning of the common exponent $\alpha = 1$ by a scale normalization condition.

(3) An honest structural simulation analysis—correcting a prior circular test—with full sensitivity analysis across distributional parameters and selection thresholds.

(4) Four substrate demonstrations spanning investment governance, clinical protocol management, software delivery standards, and lean manufacturing, including one example that clears the adoption gate after refinement.

(5) A pre-registered empirical validation path specifying exactly how the framework's predictive validity will be tested when sufficient outcome data has accumulated.

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2. Literature Review and Identification of the Gap

2.1 The Kaizen Literature: Empirical Richness, Theoretical Poverty

Imai's (1986) foundational text established Kaizen as a management philosophy centered on small, continuous improvements implemented by those closest to the process. The Toyota Production

System, developed by Ohno (1988) and Shingo (1989) between 1948 and 1975, provided its most celebrated instantiation: a system of waste elimination (muda), unevenness reduction (mura), and overburden prevention (muri) that transformed Toyota from a regional manufacturer into the world's largest automotive producer. Womack and Jones (1996) generalized these principles into Lean Thinking, extending their applicability beyond manufacturing.

The ensuing empirical literature is substantial. A 2024 systematic review covering 45 studies published between 1996 and 2024 found consistent positive outcomes across sectors (Sahmi & El Abbadi, 2024). The DORA research program (Forsgren, Humble, & Kim, 2018) documented four key DevOps metrics across 31,000 software professionals.

What is conspicuously absent is mathematical formalization. Singh and Singh (2009), in a comprehensive review of Kaizen philosophy, explicitly note the absence of formal measurement frameworks. No prior work provides a substrate-agnostic scoring function for improvement candidates that is simultaneously bounded, auditable, and self-referential.

2.2 Process Improvement Models: PDCA, PDSA, and Their Limits

Shewhart's (1939) Plan-Do-Check-Act cycle and the IHI's Plan-Do-Study-Act variant provide the iterative structure within which Kaizen operates. These models share a structural limitation: they are process frameworks, not scoring algorithms. They tell practitioners when and how to iterate; they do not tell them which improvement candidates to prioritize. Meta-Kaizen fills this gap by providing the missing selection mechanism.

2.3 Cross-Domain Inspiration: Bisociation and Innovation Theory

The theoretical foundation for cross-domain inspiration derives from Koestler's (1964) concept of bisociation: the simultaneous activation of two habitually incompatible frames of reference, producing creative synthesis at their intersection. Tversky's (1977) feature model of similarity provides the mathematical grounding for measuring domain distance via the Jaccard coefficient. March's (1991) exploration-exploitation framework establishes that organizations optimizing only exploitation suffer long-run rigidity; Meta-Kaizen's domain rotation mechanism formally operationalizes the exploration imperative.

2.4 Self-Referential and Self-Evolving Systems

Schmidhuber's (2007) Gödel Machine formalizes self-code modification guided by proof of utility improvement. These computational approaches share a limitation: they are computationally intensive, domain-specific, and incompatible with lightweight, human-auditable governance structures. Meta-Kaizen occupies the unexplored middle ground: formally self-referential but computationally trivial.

2.5 Antifragility and Adaptive Governance

Taleb's (2012) concept of antifragility—systems that gain capability from stressors—provides normative motivation. Taleb and Douady (2013) characterize antifragility as positive convexity of outcome with respect to variance. We bound this claim precisely: the antifragility property applies to the governance selection mechanism—the framework for choosing improvements becomes better as it accumulates retrospective outcome data—not to any particular substrate outcome.

3. Conceptual Foundation

3.1 The Substrate Abstraction

Meta-Kaizen operates on governance substrates: structured documents, process specifications, or policy frameworks that (a) define the operating rules or objectives of a system, (b) are revisable by identified principals, and (c) have measurable outcomes against which improvements can eventually be assessed.

Definition 3.1 (Governance Substrate)

A substrate S is a tuple $S = (D, P, \Omega, M)$ where:

D = the document set (current version and all prior versions)

P = the principal set (individuals or roles with authority to approve changes)

Ω = the outcome space (measurable variables against which improvements are assessed)

M = the metric function $M: D \times \text{time} \rightarrow \Omega$

An improvement candidate for S is a proposed revision δ to D, characterized by four properties: Novelty (domain-distance from recent candidates), Impact (projected effect on Ω), Complexity (operational burden), and Timeliness (timeframe of relevance). Only the calibration of Impact's normalizing constant I_{max} varies by substrate; the KVS formula is identical across all.

3.2 Self-Referentiality

Meta-Kaizen is self-referential in the following precise sense: the Meta-Kaizen process itself—including its domain rotation schedule, its KVS formula, its threshold parameter, and its approval gate—is a governance substrate. Let S_{MK} denote this meta-substrate. Improvement candidates for S_{MK} can be scored using KVS exactly as improvement candidates for any other substrate are scored.

Referee objection anticipated: Does self-referentiality risk infinite regress or instability? No, for two reasons. First, the principal gate prevents any self-modification from being adopted without human review. Second, the KVS threshold applies equally to meta-level improvements. The governance structure is the circuit breaker that prevents runaway self-modification. Formal treatment of temporal consistency appears in Paper 2, Theorem 7.1.

3.3 The Orthogonal Inspiration Mechanism

The domain rotation schedule must be pre-committed before the brainstorming session begins—the pre-registration equivalent in empirical research. The administrator records the domain selection, its Jaccard distance from the prior four domains, and the resulting Novelty Factor before generating any candidate ideas. The operational target is $E[N] \geq 0.70$ over a rolling 4-scan window.

4. Mathematical Formulation

This section presents the complete formal specification of Meta-Kaizen. We begin with the axiomatic derivation of the KVS functional form—answering the fundamental question of why the formula takes the form it does. The derivation proceeds in four stages: (i) from Axioms 1–4, the Luce-Tukey theorem establishes an additive representation; (ii) Axiom 5 eliminates the additive form; (iii) Axiom 6 pins equal exponents; (iv) a scale normalization condition pins the common exponent at $\alpha = 1$.

4.1 Axiomatic Derivation of the KVS Functional Form

We seek a function $f: [0,1]^4 \rightarrow [0,1]$ mapping component scores (N, I', C', T) to an improvement quality score. The form of this function is derived, not assumed.

Axiom 1 (Weak Order)

For any two improvement candidates a and b, either a is at least as good as b or b is at least as good as a, and this ordering is transitive. This is the minimum condition for any measurement to be meaningful.

Axiom 2 (Preferential Independence)

The preference ordering over candidates that differ only in component N is independent of the values of I', C', and T. This axiom holds symmetrically for all four components.

Note: This is a modeling assumption, not an empirical regularity. In practice, preferences over novelty may interact with impact levels. Violations would weaken but not destroy the derivation—they would require replacing equal weights with estimated cross-factor coefficients. The axiom is the standard conjoint measurement starting point (Luce & Tukey, 1964).

Axiom 3 (Solvability)

For any three components held fixed, there exists a value of the fourth component that achieves any target quality score in [0,1]. This ensures the scale is continuous and complete.

Axiom 4 (Archimedean Condition)

No finite sequence of equal-interval steps in any component is unbounded. This ensures no component can be infinitely more important than another and guarantees the boundedness of all standard sequences.

Axiom 5 (Essentialness with Veto Power)

A zero value on any single component renders the candidate inadoptable:
$f(0, I', C', T) = f(N, 0, C', T) = f(N, I', 0, T) = f(N, I', C', 0) = 0$
for all component values.

Defense: A candidate with $N = 0$ is something already done in exactly this form—it is a reimplementation, not an improvement. With $I' = 0$, no improvement is produced. With $C' = 0$,

implementation costs overwhelm any benefit at current organizational capacity. With $T = 0$, the relevance window has passed. In all four cases, zero behavior reflects the actual structure of improvement quality. An organization permitting impact to compensate for zero novelty is not enforcing novelty—it is making novelty advisory.

Axiom 6 (Marginal Symmetry)

The marginal rate of substitution between any two components is equal at all points where their values are equal. Formally, $\partial f/\partial N = \partial f/\partial I'$ whenever $N = I'$; and symmetrically for all pairs of components.

Stage 1: From Axioms 1–4 to the Additive Representation

Theorem (Luce & Tukey, 1964; Krantz et al., 1971, Ch. 6)

From Axioms 1–4, the fundamental theorem of additive conjoint measurement establishes that there exist scale functions φ_N, φ_I, φ_C, φ_T such that the preference ordering is represented by an additive form:

$$F = \varphi_N(N) + \varphi_I(I') + \varphi_C(C') + \varphi_T(T)$$

This representation is unique up to positive affine transformations of the φ_i functions.

Stage 2: Axiom 5 Eliminates the Additive Form

We test Axiom 5 against the additive form. Setting $N = 0$: $\varphi_N(0) + \varphi_I(I') + \varphi_C(C') + \varphi_T(T) = 0$ for all I', C', T. This requires $\varphi_I(I') + \varphi_C(C') + \varphi_T(T) = -\varphi_N(0)$ for all I', C', T—which is impossible unless all three are constants, contradicting that they vary with their arguments. The additive form is inconsistent with Axiom 5 and is eliminated.

The multiplicative form $f(N, I', C', T) = N^\alpha \times I'^\beta \times C'^\gamma \times T^\delta$, where $\alpha, \beta, \gamma, \delta > 0$, satisfies Axiom 5 immediately (any zero component produces $f = 0$) and satisfies Axioms 1–4 by construction (it is monotone, continuous, and orderable).

Stage 3: Axiom 6 Pins Equal Exponents

Under the multiplicative form, the partial derivatives are:

$$\partial f/\partial N = \alpha \times N^{(\alpha-1)} \times I'^\beta \times C'^\gamma \times T^\delta$$

$$\partial f/\partial I' = \beta \times I'^{(\beta-1)} \times N^\alpha \times C'^\gamma \times T^\delta$$

Setting $\partial f/\partial N = \partial f/\partial I'$ at the point $N = I'$:

$$\alpha \times N^{(\alpha-1)} \times I'^\beta = \beta \times I'^{(\beta-1)} \times N^\alpha$$

$$(\alpha/\beta) \times (I'/N)^{(\beta-\alpha)} = 1 \quad \text{at } N = I'$$

$$(\alpha/\beta) \times 1^{(\beta-\alpha)} = 1 \;\rightarrow\; \alpha/\beta = 1 \;\rightarrow\; \alpha = \beta$$

By the same argument applied symmetrically to all pairs of components: $\alpha = \beta = \gamma = \delta$. The multiplicative form reduces to $f = (N \times I' \times C' \times T)^\alpha$ for some $\alpha > 0$.

Stage 4: Scale Normalization Pins $\alpha = 1$

Stages 1–3 establish that the common exponent must be equal across all components but do not determine its value. The common exponent α is pinned at 1 by the following Scale Normalization Condition, which is not an additional axiom but a consequence of how the components are defined:

Scale Normalization Condition

Each component (N, I', C', T) is already normalized to $[0,1]$ with a meaningful interpretation at every point in that range: $N = 0.5$ means half-maximum domain novelty; $I' = 0.75$ means an estimated impact at 75% of the domain maximum.

In the Luce-Tukey additive representation $F = \Sigma\varphi_i(x_i)$, the scale functions φ_i are unique up to positive affine transformations. The canonical choice when components are already expressed on normalized $[0,1]$ scales with proportional interpretations is the identity: $\varphi_i(x_i) = x_i$. This reflects the principle that components are used at face value (Krantz et al., 1971, Ch. 9, Section 9.2.1).

Under the identity scale, the additive representation $F = N + I' + C' + T$. The multiplicative form $(N \times I' \times C' \times T)^\alpha$ must correspond to this representation via the order-preserving bijection $\log(f) = \alpha \times [\log(N) + \log(I') + \log(C') + \log(T)]$.

The Scale Normalization Condition selects $\alpha = 1$ as the maximally parsimonious choice, not as a uniquely forced mathematical conclusion. To be precise: any $\alpha > 0$ is consistent with Axioms 1–6 and with the $[0,1]$ normalization of components. $\alpha = 2$ is equally well-defined on $[0,1]$-scaled quantities and would apply a convex transformation to each component. The condition that selects $\alpha = 1$ is the requirement that a unit increase in any component's proportional value produces a unit increase in the proportional score—i.e., that the scoring function is linear in each normalized component at face value. Under $\alpha \neq 1$, component value x is treated as if it had weight x^α in the score, which violates the face-value interpretation of the normalized scale.

This is a modeling principle, not an axiom. It is the strongest parsimony argument for $\alpha = 1$, but it is not a logical necessity. The Corrections Register (Correction 2) states this limitation explicitly. The derivation should be read as: given the six axioms, the unique structure is $(N \times I' \times C' \times T)^\alpha$; the Scale Normalization Condition then selects $\alpha = 1$ as the canonical choice for proportionally-interpreted components.

Conclusion: Under Axioms 1–6 and the Scale Normalization Condition, the canonical form is $f(N, I', C', T) = N \times I' \times C' \times T$. □

Note: Organizations that possess retrospective calibration data suggesting that certain components have systematically larger or smaller predictive power may relax this to differential exponents ($\alpha \neq 1$, or unequal exponents by relaxing Axiom 6). The axiomatic

derivation identifies exactly which axiom or condition to revise in each case.

Theorem 4.1 (KVS Functional Form)

Under Axioms 1 through 6 and the Scale Normalization Condition, the canonical functional form representing improvement quality is:

$$KVS_j = N_j \times I'_j \times C'_j \times T_j$$

The multiplicative structure follows necessarily from Axiom 5 (unique derivation).
The equal exponents follow necessarily from Axiom 6 (unique derivation).
The common exponent $\alpha = 1$ follows from the Scale Normalization Condition (canonical modeling choice, strongest parsimony argument; see Stage 4 for the limitation statement).
The equal-weight default is the maximally conservative prior in the absence of domain-specific calibration data. □

4.2 The Kaizen Variation Score

Definition 4.1 (Kaizen Variation Score)

$KVS_j = N_j \times I'_j \times C'_j \times T_j$

where $N_j, I'_j, C'_j, T_j \in [0,1]$ are defined in Sections 4.3–4.6.
This form is derived in Section 4.1; it is not stipulated.

Property 4.1 (Boundedness). $KVS_j \in [0,1]$ for all improvement candidates j. This is a direct consequence of Definition 4.1: the product of four numbers each in [0,1] is in [0,1]. It is stated as a Property rather than a Theorem because it follows immediately from the definitions without non-trivial inference. Its significance is architectural: boundedness ensures that KVS scores are comparable across substrates and organizations without renormalization.

Property 4.2 (Monotonicity). KVS_j is non-decreasing in each of N_j, I'_j, C'_j, T_j, holding the others fixed. For any factor $f_k \in \{N_j, I'_j, C'_j, T_j\}$: $\partial KVS_j/\partial f_k = \prod_{i \neq k} f_i \geq 0$, since all factors are non-negative. Its significance is normative: it ensures that improving any component cannot decrease the overall score, providing correct incentives.

Theorem 4.2 (Self-Referential Applicability)

Let S_{MK} denote the Meta-Kaizen process itself as a substrate. Then KVS is well-defined for improvement candidates targeting S_{MK}.

Proof: S_{MK} satisfies the substrate definition $S = (D, P, \Omega, M)$:
- D = the KVS formula specification and rotation schedule
- P = the adopting organization's governance principals
- Ω = ranking fidelity of KVS against realized improvement quality
- M = the historical log of scored candidates versus outcomes

Therefore improvement candidates targeting S_{MK} have well-defined N, I′, C′, T values under their respective definitions and KVS is computable.

Note: This theorem establishes well-definedness, not convergence or optimality. It does not claim that self-referential application of KVS converges to an optimal formula; convergence is addressed in Paper 2 through the Governance Closure Theorem and its temporal resolution. □

4.3 Novelty Factor (N)

Definition 4.2 (Novelty Factor)

$N_j = 1 - \max_{\tau \in \{t-1, t-2, t-3, t-4\}} \mathrm{Jaccard}(K_j, K_\tau)$

where $\mathrm{Jaccard}(A, B) = |A \cap B| / |A \cup B|$

The maximum (not mean) Jaccard similarity is used to prevent cycling through slightly varied phrasings of the same domain. Operational target: $E[N] \geq 0.70$ over a rolling 4-scan window.

Why Jaccard? Three reasons: (1) It operates on discrete keyword sets, requiring no trained embedding model—preserving computational universality and auditability. (2) Boundedness in [0,1] follows directly from set theory. (3) For domain novelty enforcement—a structural question about whether conceptual domains overlap—keyword set intersection is more transparent than embedding-space geometry. Organizations requiring higher semantic precision may substitute cosine distance over TF-IDF vectors; the framework is agnostic to this substitution provided the metric is bounded and symmetric.

4.4 Normalized Impact (I′)

Definition 4.3 (Normalized Impact)

$I'_j = \min(1, \mu_j / I_{max})$

where μ_j is the administrator's point estimate of annualized improvement in the substrate's primary outcome metric, and I_{max} is the pre-committed maximum plausible improvement for the substrate.

Substrate Domain	Recommended I_{max}	Units
Financial governance (IPS)	3.0	pp/yr annualized return improvement
Clinical protocol	5.0	pp reduction in target adverse event rate
Software delivery (DORA)	50%	improvement in primary DORA metric
Lean manufacturing	10%	reduction in target waste category

On subjectivity: Administrator-estimated impact is real but bounded and governable. Three controls apply: (1) I_max is calibrated annually against realized historical improvements. (2) Impact estimates are logged at proposal time and compared to realized outcomes at annual review. (3) The principal gate provides an independent check. The framework makes subjectivity visible, bounded, and auditable—a substantial advance over implicit, unlogged judgment.

4.5 Normalized Inverse Complexity (C′)

Definition 4.4 (Normalized Inverse Complexity)

$C'_j = (C_{max} - C_j) / (C_{max} - C_{min}) = (2.0 - C_j) / 1.0$

where $C_j \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$

The linear rescaling maps [1.0, 2.0] bijectively onto [1.0, 0.0], fully spanning [0,1] and reaching zero at maximum complexity. This corrects the 1/C formulation used in some
prior treatments, which maps [1.0, 2.0] to [1.0, 0.5]—a compressed range that never reaches zero regardless of complexity.

Raw C	Operational Description	C′
1.0	Zero additional operational steps or resources required	1.00
1.2	One new lightweight step; under 30 minutes per quarter	0.80
1.4	Two new steps or one new data feed; under 2 hours per quarter	0.60
1.6	Moderate process redesign; involves 2–3 stakeholders	0.40
1.8	Significant systems, legal, or structural work required	0.20
2.0	Major restructuring; capital, regulatory approval, or multi-year implementation	0.00

4.6 Timeliness Score (T)

Definition 4.5 (Timeliness Score)

$T_j = \max(0, 1 - h_j / H_{max})$

where h_j = months to relevance; H_{max} = 60 (default).

H_{max} = 60 months is the default and should be revisited at annual review for substrates with materially different planning horizons (e.g., infrastructure projects with 10-year cycles may adopt H_{max} = 120 months). Any revision requires principal approval and is logged.

4.7 Approval Gate and Derived Threshold

The adoption threshold $\kappa = 0.50$ is derived from Bayesian decision theory, not stipulated. Let L_{FP} denote the loss from a false positive (adopting a poor improvement candidate) and L_{FN} the loss from a false negative (rejecting a good one). Adopt the standard Bayesian decision framework (Duda, Hart & Stork, 2001):

Theorem 4.3 (Threshold Optimality)

Setup: Let G = 'candidate is genuinely good,' P = 'candidate is genuinely poor.' Prior $\pi = P(G)$. Score $s \in [0,1]$ is a noisy signal with continuous conditional densities $p(s|G)$ and $p(s|P)$. Decision rule: adopt if $s \geq \kappa$, reject otherwise. Expected loss:

$$R(\kappa) = L_{FP} \times P(s \geq \kappa \mid P) \times (1-\pi) + L_{FN} \times P(s < \kappa \mid G) \times \pi$$
$$= L_{FP} \times (1 - F_P(\kappa)) \times (1-\pi) + L_{FN} \times F_G(\kappa) \times \pi$$

Differentiating: $dR/d\kappa = -L_{FP} \times p_P(\kappa) \times (1-\pi) + L_{FN} \times p_G(\kappa) \times \pi$

The optimal threshold κ^* satisfies the likelihood ratio condition:

$$p_G(\kappa^*) / p_P(\kappa^*) = L_{FP} \times (1-\pi) / (L_{FN} \times \pi)$$

SYMMETRIC SPECIAL CASE ($\kappa^* = 0.50$): When $L_{FP} = L_{FN}$ (symmetric losses), $\pi = 0.5$ (uniform prior over improvement quality), and the score distributions are symmetric around $\kappa = 0.5$ so that $p_G(0.5) = p_P(0.5)$, the likelihood ratio condition gives LR = 1, so $\kappa^* = 0.50$. This is the unique threshold optimal under symmetric losses with a uniform quality prior. □

ASYMMETRIC CASE: When $L_{FP} \neq L_{FN}$ or $\pi \neq 0.5$, the general formula $\kappa^* = L_{FN}/(L_{FP} + L_{FN})$ is a commonly cited approximation derived under the assumption of uniform score distributions on [0,1]. Under uniform distributions $F_G(\kappa) = F_P(\kappa) = \kappa$, the minimization of $R(\kappa)$ yields

$$\kappa^* = L_{FP}(1-\pi) / [L_{FP}(1-\pi) + L_{FN}\times\pi]$$

For $\pi = 0.5$: $\kappa^* = L_{FP} / (L_{FP} + L_{FN})$. For the inverse convention: $\kappa^* = L_{FN}/(L_{FP} + L_{FN})$. This derivation requires distributional assumptions not otherwise stated. Practitioners using asymmetric thresholds should derive κ^* from their specific loss ratio and score distributions. The complete derivation of the optimal detection threshold under asymmetric loss — connecting κ^* to the Sentinel architecture — is provided in Paper 5, Theorem 2 (Asymmetric Loss Threshold), which establishes Paper 1's $\kappa = 0.50$ as the symmetric special case of a general result.

The substrate-specific guidelines ($\kappa^* > 0.50$ for healthcare/aviation; $\kappa^* < 0.50$ for competitive markets under disruption) provide directional guidance absent full distributional knowledge.

Override protocol: The principal may override the gate in either direction. All overrides require: (1) a reason code from a pre-defined taxonomy; (2) a written rationale (minimum 50 words); (3) entry in the Master Decision Log with timestamp, override direction, and reason code; (4) review at the

subsequent annual retrospective. Override is a feature, not a failure—it transforms invisible judgment calls into auditable governance decisions.

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5. Structural Simulation Analysis

5.1 What Simulation Can and Cannot Establish

This section replaces an earlier draft's Monte Carlo analysis that (i) measured rank correlation between KVS and the arithmetic mean of its own inputs $V^* = (N + I' + C' + T)/4$, calling V^* 'an independent ground truth'—V^* is not independent—and (ii) defined the multiplicative ground truth as Q_mult = KVS.copy(), making the recall test analytically circular with 100% recall by construction. Both errors have been corrected.

The only test that would constitute genuine empirical validation is a prospective study comparing realized outcomes for KVS-selected improvements against a control condition across multiple organizations and substrates over multiple years. That data does not yet exist. Section 5.4 specifies the pre-registered empirical validation path.

What simulation can and does establish: (1) Structural consistency—that KVS ranks candidates in ways stable under plausible input distribution assumptions. (2) Decision-theoretic characterization—how the multiplicative versus additive choice affects selection under hypothetical outcome functions. (3) Sensitivity analysis—how KVS scores respond across distributional parameter variants and selection thresholds.

5.2 Simulation Specification

We draw 10,000 improvement candidates per run across 1,000 independent runs (numpy seed = 42). Two scoring rules are evaluated: $KVS = N \times I' \times C' \times T$ (multiplicative), and $ADD = (N + I' + C' + T) / 4$ (additive arithmetic mean). Both scorers rank all 10,000 candidates; top-5% recall is computed against each ground-truth outcome function from the identical candidate pool and the same 1,000 runs.

Parameter	Distribution	Rationale
N (Novelty)	Beta(2,2) on [0,1]	Symmetric; mean 0.5; moderate novelty under rotation
μ (raw impact)	Log-Normal(μ=0.6, σ=0.55)	Right-skewed; most ideas modest impact; mode ≈ 1.2 units
$I' = \min(1, \mu/I_{max})$	Derived; $I_{max} = 3.0$	Normalized; ≈18% of draws produce $I' = 1.0$
C (raw complexity)	Uniform[1.0, 2.0]	Neutral prior over operational burden
$C' = (2.0 - C)/1.0$	Derived	Uniform[0,1] by derivation
h (horizon, months)	Uniform[0, 72]	Spans beyond 60-month cutoff; ≈17% receive T = 0
$T = \max(0, 1 - h/60)$	Derived	Linear decay; truncated at zero

5.3 Results

Factor Dominance

No single factor exceeds a dominance threshold of $\rho > 0.90$. Timeliness (T) correlates with KVS at $\rho = 0.740$—substantially higher than other factors. This is a distributional artifact: approximately 17% of candidates receive $T = 0$, collapsing their KVS to zero regardless of other values, anchoring the lower end of the KVS ranking. Operational remedy: filter obsolete candidates ($T = 0$) before scoring. The equal-weight property holds for candidates with $T > 0$.

Factor	ρ with KVS	90% CI
N alone	0.258	[0.240, 0.272]
I′ alone	0.217	[0.202, 0.233]
C′ alone	0.385	[0.370, 0.400]
T alone	0.740	[0.730, 0.749]

Heterogeneity Overlap

Under the above distributions, 89.0% [90% CI: 87.2, 90.6%] of the top-5% KVS candidates also appear in the top-5% of the additive ranking. The 11% divergence consists of candidates with heterogeneous profiles that the additive rule admits (compensating strengths) but KVS excludes (Axiom 5 veto). Whether these candidates should be included or excluded is a normative question, not an empirical one. Adopting the additive rule over KVS changes approximately 1-in-9 top-tier selections.

Decision-Theoretic Comparison (Corrected)

We consider three hypothetical outcome functions: Q_mult_noisy (true quality is multiplicative, observed with log-normal noise, $\sigma = 0.30$); Q_add (true quality is additive); Q_threshold ($Q = 1$ iff all factors > 0.5, else $Q = 0$).

Outcome Function	KVS Top-5% Recall	ADD Top-5% Recall
Q_mult_noisy (multiplicative + noise)	73.8% [71.2, 76.2]	68.4% [66.0, 70.8]
Q_add (additive outcome function)	89.0% [87.2, 90.6]	100.0% (by construction)
Q_threshold (all factors > 0.5)	58.8% [56.1, 61.5]	51.2% [48.6, 54.1]

Four findings warrant discussion. Under Q_mult_noisy, KVS achieves 73.8% recall against the additive scorer's 68.4%—a 5.4 percentage point advantage with non-overlapping confidence intervals, providing the interpretive benchmark absent from prior drafts. A random top-5% selector would achieve 5% recall by construction; both scorers vastly exceed this floor. The 73.8% figure is therefore not merely above chance but above a competitive alternative. The 26.2% miss rate reflects log-normal noise at $\sigma = 0.30$, not a model failure. Under Q_threshold—the most practically informative function, formalizing that improvements require minimum thresholds across all dimensions—KVS outperforms the additive rule by 7.5 percentage points with non-overlapping confidence intervals across all thresholds and distributional variants tested.

Sensitivity Analysis: Distributional Parameters

Parameter Variant	Overlap (KVS∩ADD)	Thresh. KVS Recall	Thresh. ADD Recall
Baseline (Beta(2,2), LN(0.6,0.55))	89.0%	58.8%	51.4%
Skewed-low N (Beta(1,3))	77.2%	92.5%	75.6%
Skewed-high N (Beta(3,1))	92.0%	40.8%	37.7%
Low-impact (LN(0.2,0.55))	86.2%	77.0%	65.5%
High-impact (LN(1.2,0.55))	91.2%	47.3%	42.9%
Wide-impact spread (LN(0.6,1.10))	89.7%	65.7%	58.0%

KVS maintains its advantage over ADD under Q_threshold across all six distributional variants. Overlap remains between 77–92% across all parameter sets.

Sensitivity Analysis: Top-k Threshold Sweep

Top-k Fraction	KVS∩ADD Overlap	Thresh. KVS Recall	Thresh. ADD Recall
Top-1%	93.0%	14.9%	14.4%
Top-2%	90.5%	28.6%	26.1%
Top-5%	89.0%	58.8%	51.1%
Top-7%	88.1%	72.2%	62.7%
Top-10%	87.7%	85.0%	74.7%

5.4 Pre-Registered Empirical Validation Path

Primary hypothesis. H_1: Among improvement candidates that meet the KVS ≥ κ threshold, the Spearman rank correlation between KVS score and realized outcome improvement (measured at 12 months post-adoption) is significantly greater than zero, stratified by substrate type.

Minimum sample: n ≥ 30 adopted improvements per substrate type per organization, aggregated across ≥ 10 contributing organizations, for a minimum total of 300 outcome observations per substrate. This is achievable in high-frequency substrates within three to four years; lower-frequency substrates such as financial governance (2–3 IPS revisions per year per organization) may require a decade or more to reach the n ≥ 300 minimum.

Reporting commitment: Results will be reported regardless of direction. If KVS is not predictive of realized outcomes, the framework will be characterized as a governance tool—transparent, auditable, and consistent—rather than a scientific measurement instrument.

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6. Substrate Demonstrations

We demonstrate Meta-Kaizen across four substrates. These are worked illustrations of the framework's application mechanics, not empirical validations. A fifth example shows how a near-

miss candidate is refined to clear the adoption gate, demonstrating KVS used both diagnostically and prescriptively.

6.1 Substrate 1: Investment Policy Statement

Worked example—Domain: Mycorrhizal Networks. Improvement candidate: revise the IPS to include a correlation-aware allocation protocol, treating portfolio positions as nodes in a network where correlation represents proximity, and systematically allocating to low-correlation 'bridge' positions.

Component	Value	Computation
N (Novelty)	0.91	Jaccard distance from prior 4 domains = 0.91
I′ (Impact)	0.67	Estimated 2.0 pp/yr ÷ I_max = 3.0
C′ (Complexity)	0.60	C = 1.4; two new data feeds required
T (Timeliness)	1.00	Immediate relevance; h = 0 months
KVS	0.366	0.91 × 0.67 × 0.60 × 1.00 = 0.366

KVS = 0.366 < κ = 0.50. The candidate does not automatically clear the gate. Diagnostic: the limiting factor is C′ = 0.60. If the protocol can be simplified to a single quarterly rebalancing step (C = 1.2, C′ = 0.80), KVS rises to 0.488—within override consideration but still below threshold. This illustrates KVS used diagnostically: identifying which factor is the binding constraint and enabling targeted improvement of the proposal.

6.2 Substrate 2: Clinical Nursing Protocol

Worked example—Domain: Agricultural Crop Rotation. Improvement candidate: implement a patient assignment rotation protocol in which nurses are systematically rotated through patient acuity categories on a pre-committed schedule, preventing expertise atrophy.

Component	Value	Computation
N (Novelty)	0.88	Jaccard distance from prior 4 domains = 0.88
I′ (Impact)	0.60	Estimated 3.0 pp/yr ÷ I_max = 5.0
C′ (Complexity)	0.80	C = 1.2; one new scheduling step
T (Timeliness)	1.00	Immediately relevant
KVS	0.422	0.88 × 0.60 × 0.80 × 1.00 = 0.422

KVS = 0.422 < κ. Strengthening the impact estimate to 3.5 pp (I′ = 0.70) raises KVS to 0.493—one plausible refinement from the gate. The diagnostic question becomes: is 3.5 pp/yr reduction in adverse events defensible from analogous rotation evidence in similar units?

6.3 Substrate 3: Software Delivery Standards (DORA)

Worked example—Domain: Lean Flow Manufacturing. Improvement candidate: introduce a 'deployment pipeline value stream map' treating each pipeline stage as a manufacturing workstation, identifying and eliminating non-value-added wait states.

Component	Value	Computation
N (Novelty)	0.91	Jaccard distance from prior 4 domains = 0.91
I′ (Impact)	0.60	Estimated 30% lead-time reduction ÷ I_max = 50%
C′ (Complexity)	0.60	C = 1.4; requires VSM workshop + tooling
T (Timeliness)	1.00	Immediately relevant
KVS	0.328	0.91 × 0.60 × 0.60 × 1.00 = 0.328

KVS = 0.328 < κ. Refinement path: if the VSM registry can be automated (C → 1.0, C′ → 1.00) and the impact estimate sharpened to 40% lead-time reduction (I′ = 0.80), KVS rises to 0.728—well above threshold. The automation investment required to bring C to 1.0 is itself a candidate for KVS scoring.

6.4 Substrate 4: Lean Manufacturing Process Specifications

Worked example—Domain: Wabi-Sabi Aesthetics. Improvement candidate: revise the process specification to explicitly permit 'good enough' intermediate states in setup operations, reducing setup anxiety and enabling faster SMED cycles.

Component	Value	Computation
N (Novelty)	0.94	Jaccard distance from prior 4 domains = 0.94
I′ (Impact)	0.50	Estimated 5% OEE improvement ÷ I_max = 10%
C′ (Complexity)	1.00	C = 1.0; document change only, no new steps
T (Timeliness)	1.00	Immediately relevant
KVS	0.470	0.94 × 0.50 × 1.00 × 1.00 = 0.470

KVS = 0.470 < κ, but the margin is narrow and C′ = 1.00 means complexity is not the limiting factor. The binding constraint is impact: I′ = 0.50 requires a 5% OEE improvement estimate.

6.5 Gate-Clearing Example: Refinement of Substrate 4

This example demonstrates KVS used prescriptively rather than diagnostically. The Wabi-Sabi candidate (Section 6.4) nearly clears the gate; the question is whether the impact estimate is conservative.

Principal's inquiry at the gate

The scoring administrator reviews Substrate 4's candidate with KVS = 0.470 < 0.50.

The diagnostic question: Is 5% OEE improvement (I′ = 0.50) conservative for a

document-only change (C′ = 1.00) with high novelty (N = 0.94)?

Literature check: Shingo (1989) and Ohno (1988) both document that SMED cycle time reductions of 7–12% are achievable through specification changes that reduce setup operator anxiety and permission-seeking behavior. A 5.5% OEE improvement would be conservative by this literature's standards.

Revised estimate: I′ = 0.55 (estimated 5.5% OEE improvement ÷ I_max = 10%).
Revised KVS = 0.94 × 0.55 × 1.00 × 1.00 = 0.517 > κ = 0.50.

Decision: Surface to principal for adoption review. Note in audit log: impact estimate revised from 5.0% to 5.5% based on literature check; original conservative estimate documented.

Note on the pattern of sub-threshold examples: Sections 6.1–6.4 all produce KVS < κ. This is not a failure of the framework—it reflects that bisociative first-pass proposals from highly orthogonal domains (N ≈ 0.88–0.94) frequently require impact or complexity refinement before clearing the gate. The diagnostic function of KVS—identifying which factor is the binding constraint—is the primary value in these cases. Section 6.5 shows how the refinement process leads to gate-clearing.

6.6 Gate-Clearing Without Revision: Substrate 5 (Clinical Protocol, Established Domain)

The four primary substrate examples (Sections 6.1–6.4) all produce KVS below threshold on first pass, reflecting the natural tension between high bisociative novelty (N ≈ 0.88–0.94) and the multiplicative veto imposed by Axiom 5. Section 6.5 showed how a near-miss candidate clears the gate after a principal-reviewed impact revision. This section demonstrates a candidate that clears the gate immediately, without revision, by combining a more moderate novelty score with higher impact and lower complexity—the parameter profile characteristic of improvements drawn from adjacent rather than maximally orthogonal domains.

Worked example — Domain: Industrial Process Control (Statistical Process Control, SPC). Substrate: Clinical Nursing Protocol. Improvement candidate: introduce a run-chart monitoring protocol for nursing handover accuracy, applying Shewhart control chart logic (Western Electric rules) to flag systematic degradation in handover completeness scores before individual failures become adverse events. The domain (industrial SPC) is adjacent to the clinical quality literature rather than maximally orthogonal, resulting in lower N than the crop-rotation example (Section 6.2) but higher impact confidence due to stronger analogical precedent.

Component	Value	Computation
N (Novelty)	0.74	Jaccard distance from prior 4 domains = 0.74. SPC is adjacent to the clinical quality improvement literature (lower domain distance than Sections 6.1–6.4) but sufficiently distinct to satisfy the novelty criterion.
I′ (Impact)	0.90	Estimated 4.5 pp/yr reduction in adverse handover events ÷ I_max = 5.0. Langley et al. (2009) document 3.5–6.0 pp

		reductions from SPC-based monitoring protocols in analogous nursing contexts; 4.5 pp is conservative within this range.
C′ (Complexity)	0.80	C = 1.2; requires one new run-chart template and one quarterly review step. No new data feeds or structural redesign.
T (Timeliness)	1.00	Immediately relevant; h = 0 months.
KVS	0.533	0.74 × 0.90 × 0.80 × 1.00 = 0.533 > κ = 0.50. Gate cleared on first pass.

KVS = 0.533 > κ = 0.50. The candidate clears the gate without revision and is surfaced to the principal for adoption review.

The contrast with Sections 6.1–6.4 is instructive. The SPC candidate's lower novelty (N = 0.74 vs. 0.88–0.94) reflects its origin in an adjacent rather than maximally orthogonal domain. This illustrates an important design property of the multiplicative structure: high novelty is not required to clear the gate—it is one of four balanced factors. A candidate with moderate novelty, strong impact, and low complexity can clear the gate where a maximally novel but lower-impact candidate cannot. The diagnostic message for practitioners is symmetric: Section 6.1–6.4's sub-threshold results reflect high-novelty first-pass candidates awaiting impact or complexity refinement; Section 6.6 reflects a moderate-novelty candidate whose impact confidence is high enough to compensate. KVS is not biased toward high-novelty candidates—it enforces balance across all four dimensions.

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7. Discussion, Limitations, and Anticipatory Responses

7.1 What Meta-Kaizen Is and Is Not

Meta-Kaizen is a formal scoring and governance framework. It is not a prediction system. KVS does not predict that a given improvement will succeed; it scores the a priori quality of the improvement candidate relative to four criteria. A KVS of 0.75 means the candidate is novel, impactful, low-complexity, and timely—not that it will produce the projected outcome.

7.2 The Subjectivity Problem

The most common objection to scoring frameworks is that subjective inputs contaminate the output. Meta-Kaizen contains two subjective inputs: μ_j (impact estimate) and C_j (complexity estimate). Both are administrator judgments. The response is threefold: subjectivity is unavoidable in any a priori evaluation of unimplemented ideas; Meta-Kaizen makes subjectivity explicit, bounded, and logged rather than implicit and invisible; and the accountability structure creates statistical pressure toward calibration over time.

7.3 The Small-N Problem

Any single organization's implementation will involve a small number of improvement candidates (5–15 per quarter). This is insufficient for statistical inference about the framework's effectiveness. The planned open-source software library (Section 8) is the mechanism for enabling multi-organization replication. The validation path sample size (n ≥ 300 per substrate) is achievable in

high-frequency substrates within three to four years; lower-frequency substrates may require longer accumulation periods.

7.4 The Multiplicative Structure: Is It Justified?

This objection is answered in Section 4.1 rather than deferred. The multiplicative structure is not a design choice requiring post-hoc defense; it is derived from Axioms 1 through 5. The additive form is shown to be inconsistent with Axiom 5 by direct proof. An organization that rejects Axiom 5 should use the additive scoring variant; the library implements both.

7.5 The Validation Gap

The most consequential limitation is that this paper cannot provide empirical evidence that KVS-selected improvements outperform unselected ones. The Section 5 simulation establishes structural properties of the scoring algorithm but cannot serve as a substitute for outcome data. We state this plainly and provide the pre-registered validation design. A framework that is formal, auditable, and self-referential has real governance value independent of its predictive validity.

7.6 Independence Axiom (Axiom 2) Violations

Axiom 2 (Preferential Independence) may fail in practice. For example, a highly novel proposal (N near 1.0) from an unproven domain may be evaluated differently at high versus low impact levels—genuine interaction effects. Such violations would require relaxing to differential weights or even a non-additive-in-logs model. We note explicitly: the axiomatic derivation identifies Axiom 2 as the specific axiom to test and potentially revise. The empirical validation path (Section 5.4) should include tests of preferential independence across substrates.

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8. Software Implementation: The metakaizen Library (Specification)

8.1 Design Principles

The following section specifies the planned metakaizen Python library architecture. The library is not yet publicly released; the specification is provided to establish the intended design and implementation contract. The repository will be published on GitHub under an open license prior to or concurrent with peer review.

Design principles: Substrate agnosticism: all substrate-specific logic is isolated in adapter classes. Reproducibility: every KVS computation is logged with full input parameters. Auditability: the Master Decision Log is append-only and exports to CSV, JSON, and XLSX. Extensibility: the novelty metric, impact normalization, and complexity scale are replaceable without touching the core engine. Transparency: both KVS (multiplicative) and ADD (additive) scoring variants are implemented, enabling direct comparison.

8.2 Architecture

Layer 1 (metakaizen.core): KVSScorer implementing Definition 4.1; NoveltyScorerJaccard implementing Definition 4.2; component normalizers implementing Definitions 4.3–4.5. Both multiplicative (KVS) and additive (ADD) scoring variants.

Layer 2 (metakaizen.substrates): SubstrateAdapter base class; four concrete substrate implementations (IPS, clinical, DORA, lean manufacturing); custom adapter template.

Layer 3 (metakaizen.rotation): DomainRotationScheduler; pre-committed schedule management; Jaccard distance enforcement; E[N] monitoring.

Layer 4 (metakaizen.audit): MasterDecisionLog (append-only, timestamped); AnnualRetrospective report generator; override pattern analysis.

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9. Conclusion

Kaizen has been one of the most consequential management innovations of the twentieth century. This paper provides its first complete formal mathematical foundation. Meta-Kaizen is a formally specified, substrate-agnostic, self-referential algorithm for continuous improvement of governance substrates. Its core contribution—the Kaizen Variation Score—is a bounded, monotone, multiplicative priority function with proved properties and a complete axiomatic derivation including the pinning of equal weights and unit exponents.

The structural simulation analysis is conducted without claiming that mathematical properties of the scoring algorithm constitute evidence of predictive validity. The pre-registered validation design in Section 5.4 specifies exactly how that empirical test will be conducted. The distance between formal possibility and empirical reality is always populated by evidence. We invite the research community to help close it. Paper 5 completes the series by addressing the decision-theoretic question this paper defers: given that the Sentinel detects an approaching regime shift, when is it rational to act? The answer—derived in Paper 5's four theorems and connecting this paper's $\kappa = 0.50$ threshold to the full asymmetric cost structure—is the capstone the series was building toward.

Corrections Register

This register documents substantive changes from earlier drafts. It is retained in the published version as evidence of the framework's self-correcting methodology.

Correction 1: Circular simulation replaced (Section 5)

Prior draft measured rank correlation between KVS and $V^* = (N + I' + C' + T)/4$, calling V^* 'an independent ground truth.' V^* is not independent—it is computed from the same four inputs as KVS. An intermediate draft compounded this by defining Q_mult = KVS.copy(), making the multiplicative recall test circular (100% by construction).

Section 5 has been replaced with a three-part structural simulation analysis. Q_mult is now KVS × exp(N(0, 0.30))—multiplicative log-normal noise—producing a non-trivial recall of 73.8% [90% CI: 71.2, 76.2]. Two sensitivity analyses added.

Correction 2: Axiom 6 derivation completed; α = 1 characterized as canonical choice

(Section 4.1)

Prior draft derived equal exponents ($\alpha = \beta = \gamma = \delta$) from Axiom 6 but did not establish why the common exponent $\alpha = 1$ rather than some other positive value. The gap has been closed by adding a Scale Normalization Condition grounded in Krantz et al. (1971, Ch. 9). A further correction from v3 review: the Stage 4 derivation now explicitly states that $\alpha = 1$ is the canonical modeling choice under proportional normalization, not a uniquely forced mathematical conclusion. Any $\alpha > 0$ is axiom-consistent; $\alpha = 1$ is selected by the parsimony principle that the scoring function is linear in each normalized component at face value. Theorem 4.1 is updated to distinguish necessary derivations (multiplicative structure, equal exponents) from the canonical choice ($\alpha = 1$).

Correction 3: Theorem 4.3 asymmetric case fully characterized (Section 4.7)

Prior draft stated the asymmetric-case formula $\kappa^* = L_FN/(L_FP + L_FN)$ without proof. The complete proof has been added for the symmetric case. The asymmetric formula is now correctly characterized as requiring distributional assumptions (uniform score distributions and uniform quality prior) and is presented as an approximation with explicitly stated conditions, not a uniquely derived result.

Correction 4: Gate-clearing example added (Section 6.5)

Prior draft showed all four substrate examples with $KVS < \kappa$, demonstrating only diagnostic use. Section 6.5 adds a worked refinement of Substrate 4 showing how the principal's literature review leads to a revised impact estimate that clears the gate, demonstrating prescriptive as well as diagnostic use.

Correction 5: Axiom 2 limitation acknowledged (Section 7.6)

Prior draft listed Axiom 2 (Independence) as a modeling assumption but did not discuss what its failure would entail. Section 7.6 now explicitly names independence violation as a testable limitation of the equal-weight derivation and identifies it as a target for empirical testing in the validation path.

Correction 6: Missing reference added (Shingo, 1989)

Shingo (1989) was cited in Section 2.1 text of earlier drafts but absent from the reference list. Full citation added to References.

Correction 7: Antifragility claim scoped (Section 2.5)

The antifragility claim is now precisely bounded: it applies to the governance selection mechanism, not to any particular substrate outcome.

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